

Why $N \uparrow$ if use lump-sum rather than proportional (marginal) income taxation

Govt needs money $G = \text{Tax}$

competitive
real-
world
equilibrium
in our
model

① Use proportional income taxes: t
income effect: income $\downarrow$ ($w(1-t) < w$)

$\therefore C \downarrow \quad l \downarrow$

"normal goods"

substitution effect: return from $N \downarrow$
(opportunity cost of leisure $\downarrow$)

$l \uparrow \quad C \downarrow$

$C \downarrow$
 $l ?$
 $N ?$

This
is also
what the
social
planner
would do

② Use lump-sum taxes

income effect: income $\downarrow$ ($w - T < w$)

$\therefore C \downarrow \quad l \downarrow$

"normal goods"

$C \downarrow$
 $l \downarrow$
 $N \uparrow$

$\therefore$ under proportional taxes

$l \geq l_{\text{planner}}$

$N \leq N_{\text{planner}}$

GDP in
the real world
 $<$ GDP without
tax distortions